The symptoms and signs which constitute the carpal tunnel syndrome (CTS) result from entrapment or compression of the median nerve within the carpal tunnel. Electrodiagnostic studies may objectively document the presence of median neuropathy within the carpal tunnel and help distinguish CTS from other disorders such as cervical radiculopathy, neurogenic thoracic outlet syndrome, proximal median nerve compression syndromes, and polyneuropathy which may either mimic or occasionally coexist with CTS. Recording median nerve responses with wrist and palm stimulation allows determination of the wrist segment conduction velocity which is a more sensitive nerve conduction parameter than wrist latency measurements. Electrodiagnostic testing permits estimation of severity and relative contribution of axonal versus demyelinative nerve injury. This information can provide prognostic information and help guide therapeutic decisions. © 1995 Mark A. Ross, MD and Jun Kimura, MD. Published by John Wiley & Sons, Inc.
Key words: carpal tunnel syndrome • median neuropathy • compression neuropathy • entrapment neuropathy • focal neuropathy

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AAEM CASE REPORT #2: THE CARPAL TUNNEL SYNDROME

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Median nerve entrapment in the wrist-to-palm segment produces a clinical condition known as the carpal tunnel syndrome (CTS). Patients with CTS typically complain of numbness in any of the lateral four digits of the involved hand; however, all digits may be affected. Pain is a common symptom which usually involves the wrist and hand but may also be perceived more proximally in the forearm, arm, or shoulder. Pain and numbness are often aggravated by use of the hands, and patients characteristically awaken at night complaining of these symptoms. Sensory loss, when present, usually affects the volar and the distal dorsal aspect of the lateral three-and-one-half digits, although variations are common. More severe cases develop weakness, and advanced cases have atrophy of median-innervated thenar muscles. The present case demonstrates the utility of electrophysiologic evaluation of suspected CTS.

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CASE REPORT

Clinical History. A 26-year-old left-handed woman was referred for evaluation of suspected left CTS. Three months previously, she developed numbness involving the lateral three digits of the left hand and pain in the left wrist. At work as a registered nurse, she observed that her symptoms were exacerbated by lifting patients and writing chart notes. Driving reproduced her symptoms. She was awakened by pain in her left wrist and paresthesias involving the lateral three digits. She denied weakness or loss of sensation. Her past medical history was unremarkable. Her referring physician diagnosed CTS and requested evaluation with electrodiagnostic studies when conservative therapy with a wrist splint and nonsteroidal antiinflammatory medication failed to improve her symptoms.

Physical Examination. The general physical examination was normal. There was no weakness or atrophy of the thenar muscles. Muscle stretch reflexes were normal. Sensation to pinprick was diminished on the volar surface of the second and third digits of the left hand but was otherwise in-

tact. On her left side, Phalen's sign was present, and Tinel's sign was elicited over the median nerve at the wrist.

ELECTRODIAGNOSTIC EXAMINATION

Methods. The following is a description of the protocol used in our laboratory for evaluation of suspected CTS. Nerve conduction studies (NCS) are performed with the subject supine in a warm room. Forearm skin temperature is measured and, if below 32°C, the limb is warmed to this level, or above, before beginning the study. The filter bandpass settings are 2 Hz to 10 kHz for motor studies and 20 Hz and 2 kHz for sensory studies. For motor and sensory responses, latency is measured to the initial negative response and amplitude is measured from baseline to negative peak. Median motor and sensory fibers are routinely studied bilaterally to allow comparison of the nerve conduction parameters between the two sides.

The compound muscle action potential (CMAP) is recorded from the thenar eminence with the active recording electrode placed over the abductor pollicis brevis (APB) muscle belly, at the proximal one third of the distance between the carpometacarpal and metacarpophalangeal (MCP) joints of the thumb. The reference electrode is placed over the APB tendon at the MCP joint. The antidromic sensory nerve action potential (SNAP) is recorded from the second digit with ring electrodes around the proximal (active) and distal (reference) interphalangeal joints.

We use surface electrodes to stimulate the median nerve at the wrist, palm, elbow, and axilla in this order. The cathode is placed 2 cm distal to the anode at all stimulus sites except for palm motor stimulation which is discussed below. The wrist stimulus site is located 3 cm proximal to the distal wrist crease between the tendons of flexor carpi radialis and palmaris longus muscles. The palm stimulus site for sensory fibers is in the midpalm, 8 cm distal to the wrist stimulation site.

To stimulate median motor fibers in the palm, that is, the recurrent thenar nerve, the cathode is initially placed at the same site as for palm sensory fibers. Unlike other stimulus sites, however, the anode is directed distally toward the base of the fifth digit. This is done to avoid depolarizing the recurrent thenar nerve beneath the anode.[25] When stimulating the median nerve in the palm, one must carefully observe the nature of the thenar twitch, as stimulation of the deep branch of the ulnar nerve can also generate a motor response recorded over the thenar eminence. If an adduction thenar twitch occurs due to ulnar nerve stimulation, the stimulating electrodes are repositioned in roughly millimeter increments toward the thenar eminence until an abduction twitch is achieved, indicating median nerve stimulation. In addition to analysis of thenar twitch, the median CMAP waveforms from palm and wrist stimulation sites are also compared. A change in the CMAP area or waveform configuration between these sites indicates a different population of nerve fibers has been stimulated. This may occur for a variety of reasons summarized in Table 1.

The elbow stimulus site is on the anterior surface of the arm between the biceps tendon and the medial epicondyle directly over the brachial pulse. The axillary stimulus site is on the medial aspect of the arm approximately 20 cm proximal to the elbow stimulus site. Sensory responses are frequently absent in normal persons with axillary stimulation. Conduction velocities are calculated for the elbow-to-wrist and wrist-to-palm segments for motor and sensory nerves, and for the axillary-to-elbow motor segment and the palm-to-digit sensory segment.

Ulnar nerve conduction studies of the affected arm are always done, recording the CMAP from the abductor digiti minimi muscle and the antidromic SNAP from the fifth digit. The ulnar nerve study may provide supportive evidence of an additional or alternative condition such as ulnar mononeuropathy, C-8–T-1 radiculopathy, lower trunk plexopathy, or an underlying polyneuropathy. If abnormalities of the ulnar and median nerves are found, the peroneal motor and sural sensory nerves are studied to investigate the possibility of polyneuropathy.

For precise localization of median sensory conduction abnormalities in the wrist-to-palm segment, the "inching technique" may be used.[23] The median nerve is stimulated in 1-cm increments from the midpalm to the distal forearm and the sensory response is recorded antidromically from the second digit (Fig. 1). The zero level, marked at the distal wrist crease, represents the proximal edge of the transverse carpal ligament.[20,35] The distal edge of the transverse carpal ligament is approximately at the "−3" level. The digital potentials thus recorded normally show a linear latency change of 0.16–0.21 ms/cm as the stimulus moves from mid-palm to distal forearm.[23] In patients with CTS, a latency difference exceeding either 0.5 ms/cm or twice that of other 1-cm segments, indicates a site of focal abnormality. Limitations of the inching technique include the length of time re-

Table 1. Analysis of median nerve CMAP with wrist and palm stimulation.

CMAP area or amplitude*	Possible causes	Confirmation
Palm = wrist	1. Normally the CMAP amplitudes with palm and wrist stimulation are approximately equal.	
	2. Axon loss also causes equal amplitudes with palm and CMAP wrist stimulation.	Compare CMAP amplitudes between sides. In unilateral carpal tunnel syndrome, axon loss causes reduced amplitude on affected side. Needle exam may reveal fibrillation potentials or abnormal motor unit potentials.
Wrist > palm	1. Submaximal median nerve stimulation in palm.	Adjust location of palm stimulus or increase intensity of stimulation in palm.
	2. Coactivation of median and ulnar nerves at wrist.	Move stimulating electrodes laterally away from ulnar nerve or decrease stimulus intensity at wrist.
Palm > wrist	1. Submaximal median nerve stimulation at wrist.	Increase stimulus intensity at wrist. When technical factors (e.g., edema, scarring) at wrist prevent supramaximal nerve stimulation, the amplitude with elbow stimulation should be comparable to palm.
	2. Coactivation of the deep ulnar nerve and recurrent median nerve in the palm.	Observe nature of thenar twitch and CMAP waveform configuration. Reposition stimulating electrodes to achieve pure abduction twitch with similar CMAP waveform configuration at palm and wrist sites.
	3. True conduction block.	Pathologic conduction block is confirmed by an abrupt change in amplitude or area over a short nerve segment with incremental stimulation. Prolonged latency and/or temporal dispersion of wrist and more proximal CMAP waveforms provide supportive evidence.

**CMAP area provides the best measure of the number of muscle fibers contributing to the response. Amplitude may be used, provided CMAP waveform configuration and duration are similar at each stimulus site.*

quired for the study, the requirement for preserved median sensory responses, and achieving adequate stimulation of sensory fibers in patients with calloused hands. The inching technique is not routinely used for evaluation of suspected CTS. Rather, we use it when the clinical diagnosis of CTS is strongly suspected and routine median NCS are normal. In our experience, inching with median motor fibers is technically difficult because of lack of consistent latency changes per centimeter distal to the "−1" level.

The selection of muscles for needle examination is influenced by the clinical presentation and the nerve conduction study results. As a general rule, needle examination should include distal and proximal median-innervated muscles and non–median-innervated muscles in the C-6, C-7, and C-8 myotomes. Even if NCS verify isolated focal median nerve conduction slowing at the wrist level, it is advisable to perform needle examination of the above-mentioned non–median-innervated muscles to exclude alternative or concomitant disorders. This is necessary because NCS alone may give false-positive results suggesting CTS[34] and the symptoms of CTS and cervical radiculopathy may overlap. Furthermore, several series have shown coexistence of CTS and cervical radiculopathy.[13,42]

Nerve Conduction Studies. Nerve conduction study results are shown in Table 2. The amplitude of median CMAPs are normal bilaterally at all stimulus sites. The 4-ms left median CMAP latency with wrist stimulation was within normal absolute limits, but the 0.8-ms prolongation relative to the right side exceeded the upper limit of normal. Median SNAPs with palm stimulation were 50 μV bilaterally, but with wrist stimulation, the left SNAP amplitude was reduced by 25 μV compared to the right. The 3.3-ms left median SNAP latency with wrist stimulation was within normal absolute limits, but excessively prolonged relative to the right side. Conduction velocities of the left median nerve wrist segment were slowed to 39 m/s for motor fibers and 41 m/s for sensory fibers. An inching study revealed a SNAP latency increase exceeding 0.5 ms/cm, associated with a 50% amplitude reduc-

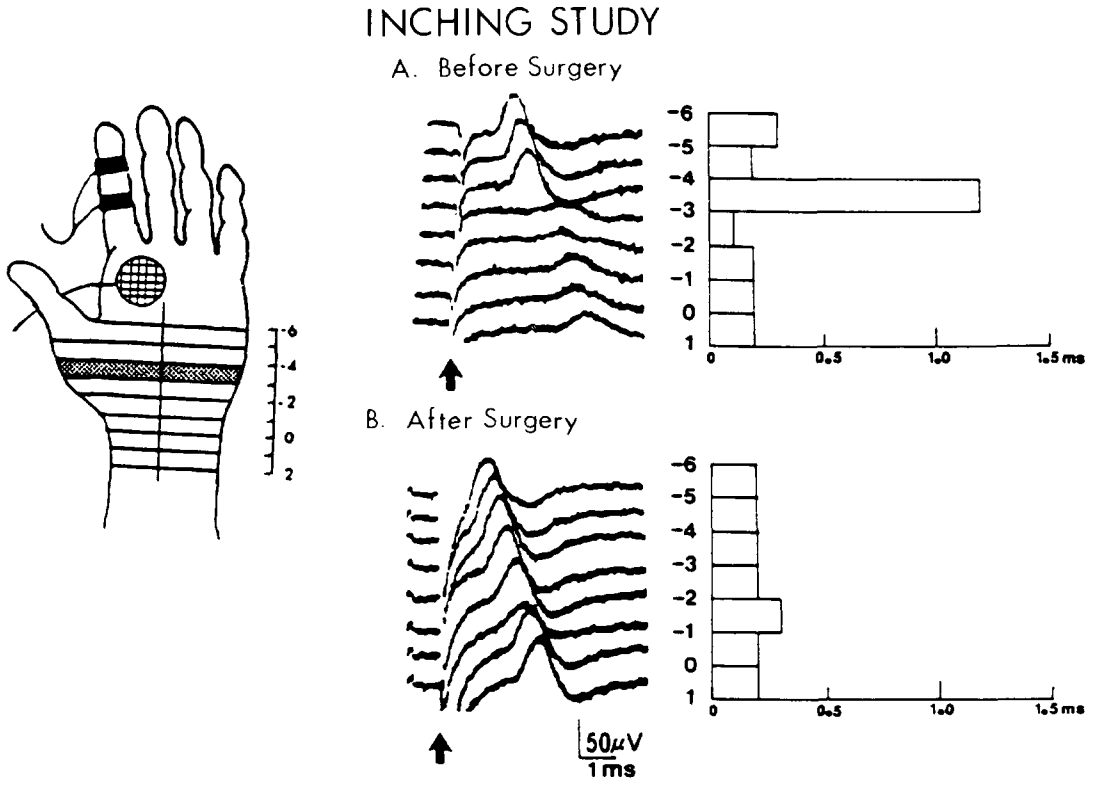


FIGURE 1. (A) Preoperative inching study. Sensory nerve action potentials (SNAPs) are recorded antidromically from the index finger with stimulation of the median nerve in 1-cm increments across the wrist. The zero level is marked at the distal wrist crease. The bar graph shows the latency difference in milliseconds between each stimulus site. Between the −3 and −4 levels an abnormally large latency difference is seen as well as a reduction in SNAP amplitude at the −3 level. **(B)** Postoperative inching study. Inching study repeated 10 months after carpal tunnel release surgery revealed normal SNAP latency difference between each stimulus site and normal SNAP amplitude at each site.

tion at the "−3" level (Fig. 1A). Nerve conduction studies of the left ulnar nerve were normal.

Needle Electromyography. Needle examinations of the left abductor pollicis brevis, first dorsal interosseous, flexor carpi radialis, and biceps muscles were performed. No fibrillation potentials were seen. Motor unit action potentials and recruitment were normal.

Interpretation. The electrodiagnostic studies revealed a left median mononeuropathy involving

Table 2. Median nerve conduction study results.

Fiber type	Stimulus site	Amplitude			Latency			Latency difference‡		Conduction velocity		
		Normal* (mV or μV)	Patient		Normal† (ms)	Patient				Normal§ (m/s)	Patient	
			Right	Left		Right	Left	Normal† (ms)	Patient		Right	Left
Motor	Palm (P)	>3.5 mV	8.0	7.5	<2.4	1.4	1.7	<0.5	0.3			
	Wrist (W)	>3.5 mV	8.0	7.5	<4.2	3.2	4.0	<0.7	0.8	W to P >38	56	39
	Elbow (E)	>3.5 mV	8.0	7.5	<8.8	6.4	7.0	<0.8	0.6	E to W >48	63	63
	Axilla (A)	>3.5 mV	8.0	7.5	<11.6	8.9	9.6	<1.1	0.7	A to E >51	72	58
Sensory												
	Palm (P)	>20 μV	50	50	<1.9	0.8	1.0	<0.4	0.2	P to D >47	75	65
	Wrist (W)	>19 μV	50	25	<3.5	2.7	3.3	<0.5	0.6	W to P >44	53	41
	Elbow (E)	>16 μV	30	15	<7.9	5.5	6.4	<0.7	0.9	E to W >53	71	65

Normative data derived from 122 nerves from 61 patients, 11–74 years of age, with no apparent peripheral nerve disease. Stimulus and recording sites are described under Methods. D = digit.
**Lower limit normal (based on distribution of normative data).*
†Upper limit of normal (mean + 2 SD).
§Lower limit of normal (mean − 2 SD).
‡Latency difference between right and left sides.

the wrist segment. The inching study precisely localized the site of abnormality to the "−3" level, corresponding anatomically to the distal edge of the transverse carpal ligament. The clinical features suggested primarily median sensory nerve involvement. This was verified by prolongation of the left median SNAP wrist latency relative to the right, focal slowing of the left median sensory conduction velocity across the wrist segment, and the inching study. The amplitude of the left median SNAP with palm stimulation was normal and equal to the unaffected side, speaking against axonal loss of sensory fibers. In contrast, the reduced amplitude median SNAP with wrist stimulation suggested conduction block of approximately 50% of left median sensory fibers. This finding, in the context of chronic stable symptoms, suggested a demyelinative nerve injury.

The normal left median CMAP amplitude and normal needle examination of median-innervated muscles argued against axonal degeneration of median motor fibers. The relatively prolonged left median motor wrist latency, and the relative focal slowing of left median motor conduction velocity across the wrist segment, suggested a partial demyelinative nerve injury of left median motor fibers within the carpal tunnel, producing slowing of conduction velocity without conduction block.

CLINICAL COURSE

Because the patient's symptoms limited her ability to work, and conservative treatment failed, she underwent surgical sectioning of the left transverse carpal ligament. When reevaluated 2 weeks after surgery, she was asymptomatic. Median nerve conduction velocity across the wrist was 30 m/s for motor fibers and 44 m/s for sensory fibers. At 10 months after surgery, she remained free of symptoms. Median nerve conduction velocity across the wrist was normal at 57 m/s for motor fibers and 62 m/s for sensory fibers. An inching study at this time was also normal (Fig. 1B).

DISCUSSION

The electrophysiology of CTS[3,4,6,10,15,21–23,27,28,29,40] has been extensively studied since the first description by Simpson in 1956.[37] Many techniques have been developed for identifying abnormal median nerve function within the carpal tunnel.[38] These include measuring distal motor and sensory latencies of the median nerve[28,39]; calculating conduction velocity across the wrist segment[4,22]; comparing the distal motor latency of the median and ulnar nerves[38]; comparing the sensory latency of the median nerve with that of the radial,[5,8,19,32] ulnar,[8,18,32] or median palmar cutaneous nerve[7]; serial recording of median SNAP latencies in 1-cm increments across the wrist[23]; comparing the median nerve terminal latency with the conduction time in the proximal segment adjusted to equal distance[26,36]; and comparing the median and ulnar orthodromic sensory latencies with digit[8,41] or palm stimulation.[9,38] Presently, there is no consensus regarding which single technique is best for detecting CTS. Studies comparing several techniques have reached different conclusions.[11,32,43] A recent review of the literature concerning electrodiagnostic studies in CTS has provided general recommendations.[1]

It is generally stated that sensory nerve testing is more sensitive than motor in CTS.[1,8,28,38] If CTS is mild, then sensory NCS will be most helpful diagnostically, as sensory fibers are typically involved earlier and to a greater extent than motor fibers. If CTS is severe enough to cause absent median SNAPs, as in a high percentage of cases in some series,[8,39] the sensory NCS do not localize the median neuropathy to the carpal tunnel, and electrodiagnosis of CTS depends upon abnormal motor conduction studies. Rarely, CTS may selectively involve motor fibers.[21,38] An anatomical basis exists for this distinctly unusual circumstance, as the motor branch exits the carpal tunnel in a separate channel.[20] Thus, median NCS for CTS should include assessment of both motor and sensory fibers. In our experience, there has been a roughly even distribution of abnormalities between sensory and motor median NCS in patients with CTS, and it is far more common to find abnormalities of both sensory and motor fibers than isolated involvement of one fiber population.

We believe that performing palm stimulation is essential. Palm stimulation increases the sensitivity of detecting CTS compared to using conventional terminal latencies from the wrist. In a series of 172 symptomatic hands previously reported from our laboratory, palm stimulation allowed detection of focal slowing of median nerve conduction velocity across the wrist in 32 hands (19%), which would have been regarded as normal by assessment of conventional terminal latencies.[23] This increased sensitivity results from separation of the relatively normally conducting palm-to-digit segment from the slowly conducting wrist-to-palm segment. Measuring conventional terminal latencies from the wrist combines these segments, allowing the normal palm-to-digit segment to mask a mild abnormality in the wrist-to-palm segment.

By comparing median sensory nerve conduction velocity in the wrist-to-palm and palm-to-digit segments, palm stimulation can also help distinguish CTS from polyneuropathy. The latter condition often has slow conduction in the palm-to-digit segment, whereas this segment usually conducts normally in CTS. Finally, including palm stimulation allows detection of conduction block in the wrist-to-palm segment. Conduction block is suspected when the CMAP amplitude with palm stimulation exceeds that of wrist stimulation. A proximal-to-distal CMAP amplitude ratio of less than 0.7 has generally been said to indicate conduction block.[2] Our laboratory uses a ratio of less than or equal to 0.5 which is a conservative figure. Lesser degrees of block may be accepted as abnormal if accompanied by other features suggesting demyelinative pathology, such as prolonged latency or temporal dispersion. Conduction block is confirmed by finding an abrupt change in amplitude or area over a short nerve segment with incremental stimulation. Conduction block of the median motor fibers at the wrist should never be diagnosed unless inadvertent coactivation of the deep branch of the ulnar nerve in the palm has been excluded. Conduction block of sensory fibers may be more difficult to confirm because physiologic temporal dispersion results in phase cancellation of some sensory fiber potentials and, consequently, a normal decline in amplitude of antidromic SNAPs with more proximal stimulation.[24] Nevertheless, the amplitude of median SNAPs with palm and wrist stimulation is comparable in normal subjects,[22,23] and a reduction of wrist SNAP amplitude by 50% or more compared to the palm SNAP suggests conduction block,[16] provided technical factors can be excluded.

Reduction of median motor or sensory response amplitudes without evidence of conduction block or temporal dispersion suggests that axon loss has occurred. Significant axon loss may occur despite preservation of amplitude values well above the lower limit of normal. Thus, in unilateral CTS, the median CMAP and SNAP amplitude values should be compared with the uninvolved hand.

Electrodiagnostic studies give an estimation of the severity of the median neuropathy and the contribution of axonal versus demyelinative nerve injury. This information can be used to influence therapeutic decisions. A demyelinative nerve injury generally implies a good prognosis for recovery,[31] providing the underlying condition causing neurapraxia can be alleviated. In contrast, electrodiagnostic evidence of axonal loss suggests a more advanced nerve injury, and is often viewed as an indication for surgical intervention. Thenar atrophy and electrodiagnostic evidence of axonal injury can suggest a relatively poor prognosis,[30] however, many patients with these features improve with surgery.[17,33] Following surgery for CTS, electrodiagnostic studies typically improve,[12,14,17,29] although complete normalization does not always occur.[29]

We believe electrodiagnostic studies are essential for evaluation of CTS. Documenting median neuropathy within the carpal tunnel and excluding an alternative or concomitant disorder should always be done before contemplating surgical therapy.

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